

李晟睿 Chengrui Li

📅 Last updated: 2026-9
✉ cnlichengrui@{meta,foxmail,gmail}.com
🌐 <https://JerrySoybean.github.io>
🎓 <https://scholar.google.com/citations?user=SXR3RXIAAAJ>
🐙 <https://github.com/JerrySoybean>
🌐 <https://www.linkedin.com/in/cnlichengrui/?locale=en-US>



Research Interests

- Computational neuroscience, neural latent variable models, statistical machine learning.
- Eye tracking experiments and data analysis.
- Fractional calculus signal processing.

Education

Georgia Institute of Technology (Georgia Tech) 2021.08 – 2026.02
Ph.D. in Computational Science & Engineering *Atlanta, USA*
M.Sc. in Mathematics (awarded in 2023.12)

- Got the offer in 2020.08 but deferred to 2021.08 due to COVID-19.
- GPA: **4.00/4.00**
- Advisor: Prof. Anqi Wu
- Thesis title: *Latent variable models and inferencing methods for understanding neural connectivity and latent dynamics*

University of Tennessee, Knoxville (UTK) 2018.08 – 2019.05
Undergraduate Non-Degree Exchange Program *Knoxville, USA*

- GPA: **4.00/4.00**

Sichuan University (SCU) 2016.09 – 2020.06
B.Eng. in Software Engineering (Computational Biology) *Chengdu, China*
B.Sc. in Biological Sciences

- GPA: **3.94/4.00**; Rank: 1/28; Wu Yuzhang Honored degree
- Advisor: Prof. Yifei Pu & Prof. Wei Deng
- Thesis title: *The application of fractional order image enhancement in computational neuroscience*

Employment

Meta Platforms, Inc. 2025.06 – present
Research Scientist, Neural Interfaces: ML *New York City, USA*
ML Algorithm team, CTRL-Labs, Reality Labs

- Meta's sEMG neural band that controls the AI display glasses.
- Multi-device transfer learning.
- Finger gesture classification, handwriting classification, and other downstream tasks.
- Attention mechanism, multi-modal fusion, unified neural latent representation.
- Offline metrics evaluation, failure modes analysis and improvement.
- Data integration experiment, loss function design, model iteration and release leadership.
- Intern mentorship on a project of label quality investigation and annotation improvement via non-causal models and auxiliary devices.
- LLM agent (Claude) development and workflow automation.

Emory University 2026.01 – present
Postdoctoral Research Consultant (Remote), Dr. Dieter Jaeger's Lab *Atlanta, USA*

- Mouse voltage imaging data analysis in a decision liking task.
- Methods: Independent component analysis, frequency band, latent disentanglement, etc.

- LLM agents for automated research and result visualization.

Meta Platforms, Inc.

2024.05 – 2024.08

Research Intern, Neuromotor Interfaces: Computational Modeling

New York City, USA

EMG Foundations team, CTRL-Labs, Reality Labs

- Meta’s sEMG neural band that controls the AI display glasses.
- Handwriting classification neural network model architecture.
- Separable convolution, S4D, state-space model, etc.
- Achieved 60% on-device inference latency reduction.

Honors and Awards

Runner-up Poster Award in the Neuro Next Initiative Launch Event , Georgia Tech.	2023
Outstanding Undergraduate Thesis , SCU.	2020
Outstanding Graduates , SCU.	2020
Finalist (<0.3%) + Frank Giordano Award (the only 1 out of 14,000), Mathematical Contest in Modeling (MCM/ICM), Consortium for Mathematics and Its Applications.	2019
“Tang Lixin” Lifetime Scholarship , Tang Lixin Education Development Foundation.	2018
First Grade Scholarship , SCU.	2018
National Scholarship , Ministry of Education of the People’s Republic of China.	2017
First Grade College Mathematical Contest , SCU.	2017

Research Publications

Peer Reviewed Conference Proceedings

- C10** “A Hitchhiker’s Guide to Poisson Gradient Estimation”
Michael Ibrahim*, Hanqi Zhao*, Eli Zachary Sennesh, Zhi Li, **Chengrui Li**[†], and Hadi Vafaii[†]
(*co-first authorship, [†]co-senior authorship)
The Forty-third International Conference on Machine Learning (ICML), 2026
- C9** “A Factorized Low-Rank RNN Framework for Uncovering Independent Neural Latent Dynamics and Connectivity”
Chengrui Li, Yunmiao Wang, Yule Wang, Weihan Li, Dieter Jaeger, and Anqi Wu
The Forty-third International Conference on Machine Learning (ICML), 2026 [**Spotlight: 2.2%**]
- C8** “Uncovering Semantic Selectivity of Latent Groups in Higher Visual Cortex with Mutual Information-Guided Diffusion”
Yule Wang, Joseph Yu, **Chengrui Li**, Weihan Li, and Anqi Wu
The Fourteenth International Conference on Learning Representations (ICLR), 2026
- C7** “Learning Time-Varying Multi-Region Brain Communications via Scalable Markovian Gaussian Processes”
Weihan Li, Yule Wang, **Chengrui Li**, and Anqi Wu
The Forty-Second International Conference on Machine Learning (ICML), 2025 [**Oral: 1.0%**]
- C6** “Exploring Behavior-Relevant and Disentangled Neural Dynamics with Generative Diffusion Models”
Yule Wang, **Chengrui Li**, Weihan Li, and Anqi Wu
Advances in Neural Information Processing Systems 36 (NeurIPS), 2024
- C5** “A Differentiable Partially Observable Generalized Linear Model with Forward-Backward Message Passing”
Chengrui Li, Weihan Li, Yule Wang, and Anqi Wu
The Forty-first International Conference on Machine Learning (ICML), 2024
- C4** “Multi-Region Markovian Gaussian Process: An Efficient Method to Discover Directional Interactions Across Multiple Brain Regions”
Weihan Li, **Chengrui Li**, Yule Wang, and Anqi Wu
The Forty-first International Conference on Machine Learning (ICML), 2024

- C3** “Forward χ^2 Divergence Based Variational Importance Sampling”
Chengrui Li, Yule Wang, Weihai Li, and Anqi Wu
The Twelfth International Conference on Learning Representations (ICLR), 2024 [**Spotlight: 5%**]
- C2** “One-hot Generalized Linear Model for Switching Brain State Discovery”
Chengrui Li, Soon Ho Kim, Chris Rodgers, Hannah Choi, and Anqi Wu
The Twelfth International Conference on Learning Representations (ICLR), 2024
- C1** “Extraction and Recovery of Spatio-Temporal Structure in Latent Dynamics Alignment with Diffusion Model”
Yule Wang, Zijing Wu, **Chengrui Li**, and Anqi Wu
Advances in Neural Information Processing Systems 35 (NeurIPS), 2023 [**Spotlight: 3%**]

Journal Articles

- J3** “Feature Reconstruction Guided Fusion Network for Hyperspectral and LiDAR Classification”
Zhi Li, Ke Zheng, Lianru Gao, Nannan Zi, and **Chengrui Li**
IEEE Transactions on Geoscience and Remote Sensing, 2025
- J2** “Cross Semantic Heterogeneous Modeling Network for Hyperspectral Image Classification”
Zhi Li, Ke Zheng, Jiaxin Li, **Chengrui Li**, and Lianru Gao
IEEE Transactions on Geoscience and Remote Sensing, 2024
- J1** “Inverse Kernel Decomposition”
Chengrui Li and Anqi Wu
Transactions on Machine Learning Research (TMLR), 2024

Abstracts

- A2** “Disentangling time-varying neural latent dynamics and circuit-level connectivity with subspace-attention gating”
Chengrui Li, Kai Lu, Yen-Shuo Su, Jingyang Ke, Chris Rodgers, Robert C. Liu, and Anqi Wu
Society for Neuroscience (SfN), 2026
- A1** “Similarity of Memory Representations Modulate Saccade Curvatures”
Golnaz Forouzandehfar, **Chengrui Li**, Aaron T. Buss, and A. Caglar Tas
Journal of Vision, 2024

Preprints

- P2** “A Revisit of Total Correlation in Disentangled Variational Auto-Encoder with Partial Disentanglement”
Chengrui Li, Yunmiao Wang, Yule Wang, Weihai Li, Dieter Jaeger, and Anqi Wu
arXiv:2502.02279, 2025
- P1** “Continuous-time systems for solving 0-1 integer linear programming feasibility problems”
Chengrui Li and Bruce J. MacLennan
arXiv:1905.04612, 2019

Invited Talks and Other Presentations

Invited Talks

- School of Life Sciences, Tsinghua University, Beijing, China** 2026.04
“Disentangling Structured, Time-Varying Neural Latent Dynamics and Their Circuit-Level Connectivity”
- Data-Driven and Large-Scale Modeling in Neuroscience Workshop, Georgia Tech** 2025.04
“One-hot Generalized Linear Model for Switching Brain State Discovery (One-hot HMM-GLM GitHub Repository on Cybershuttle)”
- School of Life Sciences, Tsinghua University, Beijing, China** 2024.08
“Latent Variable Models and Inferencing Methods for Understanding Neural Functional Connectivity”
- Affiliated Mental Health Center, Zhejiang University School of Medicine (Hangzhou Seventh People’s Hospital), Hangzhou, China** 2023.08

“Latent Variable Models for Neural Spike Train Data”

Conference Presentations

- Poster, COSYNE 2026, Lisbon, Portugal** 2026.03
“A disentangled low-rank RNN framework for uncovering neural connectivity and dynamics”
“Uncovering semantic selectivity of latent groups in higher visual cortex with diffusion models”
“Learning time-varying multi-region brain communications via scalable Markovian Gaussian processes”
- Poster, COSYNE 2025, Montreal, Canada** 2025.03
“Disentangling partially independent low-rank neural representations and connectivity”
- Poster, COSYNE 2024, Lisbon, Portugal** 2024.02
“One-hot Generalized Linear Model for Switching Brain State Discovery”
“Extraction and recovery of spatio-temporal structure in neural alignment via diffusion models”
- Poster, Neuro Next Initiative Launch Event, Georgia Tech** 2023.10
“One-hot Generalized Linear Model for Switching Brain State Discovery”
- Oral, The 11th International Conference on Brain Informatics (BI 2018), Arlington, TX, USA** 2018.12
“The Power and Beauty of Mathematics: A Prospect of Nature Inspiration & Computational Model from the Interdisciplinary View”

Miscellaneous Experience

Teaching Experiences

- Grader.** MATH 6647 Numerical Dynamical System @ Georgia Tech. Spring 2025
- Teaching assistant.** CSE 6740 Computational Data Analysis @ Georgia Tech. Fall 2023
- Teaching assistant.** CSE 6740 Computational Data Analysis @ Georgia Tech. Spring 2023
- Graduate student research mentor.** SURE @ Georgia Tech. Summer 2022

Summer Schools

- CNeuro 2020: Theoretical and Computational Neuroscience Summer School,** Tsinghua University. 2020.08
- The Chinese University of Hong Kong Summer Workshop,** Hong Kong SAR, China. 2018.08
- Cognitive Neuroscience Summer School,** Peking University. 2018.07

Other Research Experiences

- Eye tracking experiment: Response Selection** 2018.09 – 2019.10
- Designed the eye-tracking program by Experiment Builder. Completed an eye-tracking data analysis program in MATLAB. Used the eye-tracking technique to investigate the influence of different stimulus-response conditions on the eye movement trajectories.
- Web project development** 2019.07
- Developed a web project for an e-commerce platform under the SSM framework. Java + MySQL + JSP was used for full-stack agile development.
- Clinical internship at the State Key Laboratory of Biotherapy, West China Hospital, SCU** 2018.02
- Conducted tests including the Mini-Mental State Examination (MMSE), the Montreal Cognitive Assessment (MoCA), the Hamilton Anxiety Rating Scale (HAM-A), and the Hamilton Depression Rating Scale (HAM-D).

Academic Services

Area Chair

- KDD AI4Sciences Track: 2026 (February), 2027 (Cycle 1)

Reviewer

- NeurIPS: 2023–2026

- ICML: 2023–2026
- ICLR: 2024–2026
- AAAI: 2025–2027
- AISTATS: 2025–2026
- TMLR: 2025–2026
- COSYNE: 2026
- Nature Communications: 2026

Skills

- **Proficient:** Python, PyTorch, Claude Code, Codex, Cursor, Git, MATLAB, LaTeX, Mathematica, Adobe Illustrator, Experiment Builder (eye-tracking)
- **Familiar:** C/C++ , Java, SQL, Julia, R, Cinema 4D, Adobe Premiere Pro, Adobe Audition, Linux

Hobbies

Violin

New Amsterdam Symphony Orchestra, New York City
 Georgia Tech Symphony Orchestra

2025.10 – present
 2023–2025

Magic

- Performing since 2009
- Member, International Brotherhood of Magicians (I.B.M.)
- Member, Society of American Magicians

Others

- Piano, Chinese orchestra conducting, YOYO ball, aerial photography, and film/audio post-processing